

Dexter Fernandes

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https://github.com/Dexter-Fernandes

Software Engineer (Computer Vision)

Software engineer with 3+ years building real-time computer vision for 24/7 field operation, from C++17 edge inference to ML microservices deployed on Azure. Production experience with NVIDIA DeepStream video pipelines, PostgreSQL schema design, and diagnosing in-service systems. Comfortable owning components end to end: (model) design and deployment, code review, testing. University of Bristol. Based in Bristol.

WORK EXPERIENCE

GrowthStage • Remote

05/2026 - Present

Recruiter

- Evaluated computer vision and SE candidates on their technical depth across SE practices, model deployment, and MLOps practices, maintaining direct exposure to current production engineering standards.

Taco Bell • Bristol, England

09/2024 - Present

Service Champion / Team Trainer

- Progressed to Team Trainer in a fast-paced customer-facing environment, onboarding new staff and handling customer escalations.

Esbaar • Muscat, Oman

01/2023 - 08/2023

AI Engineer

- Owned the end-to-end NVIDIA DeepStream 6.x video inference pipeline for 24/7 operation on British Petroleum rigs that included model training and production debugging in harsh desert conditions.
- Hardened deployed AI services with Dockerised microservices, Kafka messaging and structured logging, reducing live video pipeline crashes by 33%.
- Led peer code reviews and introduced pytest and unittest coverage with linting into CI/CD validation workflows for edge inference modules, improving release consistency across hardware targets.
- Integrated AI modules into production systems, collaborating with cross-functional engineering and client teams to deliver fault-tolerant edge deployments 500 mile away.
- Developed a watchdog application that autonomously detected and restarted failed services on edge devices, enhancing system recoverability without site visits.
- Developed multimodal data ingestion pipelines processing 0.3 M data points and multiple video streams daily, enhancing data accessibility and analysis efficiency using Airflow and Pandas.

LivNSense Technologies • Bengaluru, India

05/2020 - 12/2022

Lead Software Engineer - AI

- Led a 4-engineer team to build a multi-stage detection and helmet-colour classification pipeline for industrial safety and inspection in TensorFlow, reducing false positives by 15% for Austin Industries and John Deere.
- Deployed and maintained containerised ML inference microservices on Microsoft Azure with REST endpoints and queue-based async processing, owning autoscaling, monitoring and cost per inference across CI/CD and RBAC.
- Built low-latency pipelines in C++17 with OpenVINO on edge hw, raising throughput from 5 to 15 FPS, and tuned camera and ISP settings on RGB and infrared streams to hold detection reliable across 24/7 operation.
- Designed the PostgreSQL schema that routed real-time video incidents into downstream analytics, supporting worker heat maps, geofencing and client-facing incident feeds.
- Defined and implemented the model optimisation workflow (QAT, pruning, fusion, FP16/INT8) to cut model size by 50% while meeting accuracy, latency and memory targets.

- Mentored junior engineers and collaborated with hardware and QA teams to integrate perception modules into deployed systems with improved delivery consistency.

PROJECTS

DeepStream RTSP Pipeline

06/2026

- Built a multi-stream NVIDIA DeepStream 9.0 / GStreamer RTSP video analytics pipeline running YOLO detection with NvDCF tracking, metadata extraction, anonymisation and re-streamed output across 3 concurrent camera feeds.
- Optimised a YOLO26n TensorRT FP16 detector on 6GB Turing hardware using dynamic-batch engine conversion and p99 latency profiling, sustaining 29.7 FPS per stream live and 131.3 FPS unthrottled, validated FP16 against FP32 for accuracy.
- Implemented a C++/CUDA TensorRT plugin (IPluginV2DynamicExt) moving post-processing into the inference engine.
- Hardened with GPU smoke tests, model-promotion gates, structured JSON logging and per-camera health metrics covering liveness, dropped frames and reconnect events.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

09/2023 - 09/2024

- **Dissertation** Radiological SLAM with LiDAR Odometry: Designed and implemented a real-time SLAM system using RTAB-Map on a Clearpath Husky, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Coursework across computer vision, sensor fusion and real-time robotics software, with lab work on physical hardware under live sensor constraints.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

07/2016 - 05/2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

SKILLS

- **Programming & Tooling:** Bash, C++17, CUDA C++, Git, linting, Linux, NumPy, OpenCV, Pandas, PostgreSQL, pytest, Python, PyTorch, Scikit-image, Scikit-Learn, TensorFlow, unittest
- **Cloud & MLOps:** Apache Airflow, Apache Kafka, API design, CI/CD, containerised microservices, Docker, DVC (Data Version Control), Microsoft Azure, Microsoft Azure, MLflow, Production monitoring, REST APIs, Structured logging
- **Computer Vision Models:** ByteTrack, EfficientDet, Instance Segmentation, NvDCF, Object detection, Object tracking, RT-DETR, YOLO
- **Datasets, Annotation & Evaluation:** Confusion-matrix-driven failure clustering, CVAT, Dataset versioning, GRAD-CAM, Synthetic data generation, TIDE (Toolbox for Identifying Detection Errors)
- **Edge Devices:** CUDA, GStreamer, latency profiling, NVIDIA DeepStream, OpenVINO, TensorRT